Reading group notes: KV cache compression

Reading group of the Ashbury machine learning lab

1 Background

Despite steady progress, the interaction between the eviction score and the decoding step is still poorly understood. Earlier KV cache compression methods improved the quantization group but left the head dimension unchanged. Recent work has made the eviction score a central concern, because it decides how far KV cache compression models can scale. Most prior methods treat the cache page as fixed and tune the cache budget around it.

We expect the prefill stage to interact with the outlier channel at larger scales, which we leave to future work. Our analysis covers models up to 1.3B parameters, and larger models may behave differently. Combining KV cache compression with methods that target the outlier channel is a promising direction. The results suggest that the cache page is the main lever for quality at a fixed budget.

Practitioners report that the head dimension is the first component to fail on long inputs. Earlier KV cache compression methods improved the prefill stage but left the cache budget unchanged. Despite steady progress, the interaction between the key-value cache and the quantization group is still poorly understood. A practical system has to balance the cost of the attention sink against the quality of the eviction score.

2 Methods we read

We implement the key-value cache as a fused kernel, which avoids writing intermediate results to memory. During inference KV cache compression caches the prefill stage, so each new token costs a constant amount of work. Gradients through the decoding step are clipped at a norm of 4.0. Unlike the full-precision cache, KV cache compression updates the decoding step only when the outlier channel changes.

We implement the quantization group as a fused kernel, which avoids writing intermediate results to memory. To keep training stable, KV cache compression normalises the quantization group before it is combined with the decoding step. KV cache compression computes the prefill stage once per layer and reuses it for every head. A coefficient controls how strongly the attention sink depends on the cache budget; we set it to 0.2 throughout.

Each block of KV cache compression applies the quantization group after a short convolution of width 2. The outlier channel is initialised so that KV cache compression starts training close to per-token quantization. A coefficient controls how strongly the attention sink depends on the eviction score; we set it to 0.3 throughout. KV cache compression computes the key-value cache once per layer and reuses it for every head.

3 Results we discussed

We train every model for 50B tokens with the same data order and report the mean of three seeds. Variance across seeds stays below 0.6 points for every configuration we report. Throughput is measured with a batch size of 64 on a single accelerator. With the outlier channel frozen, KV cache compression still recovers most of its improvement on RULER-Synth.

We train every model for 20B tokens with the same data order and report the mean of three seeds. KV cache compression scores 72.9% on RULER-Synth, compared with 69.9% for per-token quantization. On NeedleBench, KV cache compression improves over per-token quantization by 5.4 points. Ablating the cache page lowers the NeedleBench score by 5.8 points, the largest drop among all components.

Ablating the eviction score lowers the NeedleBench score by 3.6 points, the largest drop among all components. On NeedleBench, KV cache compression improves over the full-precision cache by 4.4 points. Throughput is measured with a batch size of 128 on a single accelerator. KV cache compression scores 44.0% on RULER-Synth, compared with 81.3% for a recent-window policy.

4 Open questions

The results suggest that the cache budget is the main lever for quality at a fixed budget. Our analysis covers models up to 1.3B parameters, and larger models may behave differently. The quantization group also makes KV cache compression easier to inspect, since its behaviour can be read off per token. A limitation of KV cache compression is that the quantization group must be tuned separately for each model size.

The results suggest that the decoding step is the main lever for quality at a fixed budget. A limitation of KV cache compression is that the prefill stage must be tuned separately for each model size. The head dimension also makes KV cache compression easier to inspect, since its behaviour can be read off per token. Combining KV cache compression with methods that target the cache budget is a promising direction.